

Question 2

[5 points]

Prove that the language $A_2 = \{0^n 1^m 0^n \mid m \neq n\}$ is not a regular language.

$S =$ in terms of p

$$0^p 1^{p+1} 0^p \quad 0^p 1^{p-1} 0^p$$

$|S| \geq p \rightarrow p \quad p+1 \quad p = 3p+1 \geq p$

$x = 0^a \quad y = 0^{p-a} \quad z = 1^{p+1} 0^p$ $a < p$

$$|xy| \leq p \rightarrow |0^a 0^{p-a}| = |0^{p-a+a}| = |0^p| = p \leq p \checkmark$$

$$|y| = |0^{p-a}| > 0$$

Pump down $= \frac{0^a}{x} \frac{1^{p+1} 0^p}{z} = \underline{0^a} \underline{1^{p+1} 0^p}$

Pump up $= \frac{0^a}{x} \frac{0^{p-a}}{y} \frac{0^{p-a}}{y} \frac{1^{p+1} 0^p}{z} = 0^{a+p-a+p-a} 1^{p+1} 0^p$

$$= 0^{2p-a} 1^{p+1} 0^p$$

PL fails here
 → proof by contradiction shows that it's not a RL.

Prove that the following languages are not regular. You may use the pumping lemma and the closure of the class of regular languages under union, intersection, and complement.

- $\{0^n 1^m 0^n \mid m, n \geq 0\}$
- $\{0^m 1^n \mid m \neq n\}$
- $\{w \mid w \in \{0,1\}^* \text{ is not a palindrome}\}^8$
- $\{wtw \mid w, t \in \{0,1\}^*\}$

$\{0^i 1^j 0^k \mid i=j \text{ or } j=k\}$ show it's not regular using PL

$\{00110\}$

$$0^p 1^p 0^{p+1} = 0^a 0^{p-a} 1^p 0^{p+1}$$

$$0^{p+1} 1^p 0^p = 0^a 0^{p-a-1} 0 1^p 0^{p+1}$$

$\rightarrow$ $\underline{00110}$
 $\hookrightarrow$ 01100

$$\begin{aligned}
 0^a 1^p 0^p &= 0^a 0^{p-a-1} 0 1^p 0^{p+1} \\
 0^p 1^p 0^p &= 0^a 0^{p-a} 1^p 0^p \\
 &\quad x \quad y \quad z
 \end{aligned}$$

$|xy| \leq p$

$0^a 0^{p-a} 0^{p-a} 1^p$ pump up
 $0^{2p-a} 1^p 0^p$

$$\begin{aligned}
 &0^a 0^{p-a} 0^{p-a} 1^p 0^{p+1} \\
 &0^{p+p-a} 1^p 0^{p+1} \\
 &0^{2p-a} 1^p 0^{p+1}
 \end{aligned}$$